

CAREFLOW DEMO GUIDELINE: ADULT TYPE 2 DIABETES MANAGEMENT

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SYNTHETIC DEMONSTRATION DOCUMENT. This guideline was written for the CareFlow AI portfolio project. It is fictional, simplified and must not be used as real clinical guidance.

1. Purpose and Scope

This guideline describes how CareFlow General Hospital (a fictional organisation) manages adults with type 2 diabetes mellitus (T2DM) in outpatient clinics and on inpatient wards. It applies to doctors, nurses and pharmacists. It does not apply to type 1 diabetes, gestational diabetes or patients under 18 years, who follow separate protocols.

2. Diagnosis and Classification

T2DM is confirmed when HbA1c is 6.5% or higher on two occasions, or fasting plasma glucose is 126 mg/dL or higher on two occasions, or a random glucose of 200 mg/dL or higher is recorded with classic symptoms. Every new diagnosis must be coded with the correct ICD-10 code (E11 family) and recorded in the problem list.

3. Glycemic Targets

3.1 General adult target

For most non-pregnant adults the HbA1c target is below 7.0% (53 mmol/mol). A lower target of 6.5% may be considered for younger patients with short disease duration when it can be achieved without hypoglycemia.

3.2 Individualised targets for older or comorbid patients

A less stringent target of below 8.0% is appropriate for adults aged 65 or older with multiple chronic conditions, chronic kidney disease stage 3 or worse, a history of severe hypoglycemia, or limited life expectancy. HbA1c values below 6.5% should be avoided in frail older adults because the risk of hypoglycemia outweighs the benefit.

4. Monitoring

4.1 HbA1c monitoring

Measure HbA1c every 3 months in patients who are not at target or whose therapy has changed in the last 3 months. Patients who are stable at target may be tested every 6 months. An HbA1c rise of more than 0.5% between two consecutive tests should prompt a review of adherence, diet and treatment within 4 weeks.

4.2 Kidney function monitoring

Measure eGFR and the urine albumin-creatinine ratio (UACR) at least once a year in every patient with T2DM. Increase the frequency to every 3 to 6 months when eGFR is below 60 mL/min/1.73m² or UACR is above 30 mg/g. A sustained fall in eGFR of more than 25% requires a medication review, because metformin, SGLT2 inhibitors and DPP-4 inhibitors all have renal dosing rules.

4.3 Cardiovascular risk monitoring

Check blood pressure at every visit and a lipid profile (including LDL cholesterol) at least annually. Most patients aged 40 to 75 years should receive a moderate-intensity statin.

4.4 Complication screening

Perform a comprehensive foot examination at least once a year and at every visit when neuropathy is present. Arrange retinal screening every 1 to 2 years. Screen for depression annually.

4.5 Self-monitoring of blood glucose

Patients on insulin must self-monitor blood glucose (SMBG) at least once daily, usually fasting. Patients on basal-bolus insulin should test before meals and at bedtime. Patients on oral agents alone do not routinely need SMBG unless they take a sulfonylurea.

5. Pharmacological Therapy

5.1 Metformin

Metformin is the first-line agent for most patients. Start with 500 mg once or twice daily with meals and titrate every 1 to 2 weeks to a usual maintenance dose of 1000 mg twice daily. Renal dosing: when eGFR is between 30 and 44 mL/min/1.73m² the total daily dose must not exceed 1000 mg (for example 500 mg twice daily), and metformin should not be newly started in this range. Metformin is contraindicated when eGFR is below 30. Hold metformin before iodinated contrast procedures in patients with eGFR below 60 and during acute illness with dehydration.

5.2 SGLT2 inhibitors

An SGLT2 inhibitor (for example empagliflozin 10 mg once daily) is recommended for patients with chronic kidney disease (eGFR 20 or above) or heart failure, independently of HbA1c, because of kidney and heart protection. The glucose-lowering effect is reduced when eGFR is below 45, so an additional agent may be required for glycemic control. Teach sick-day rules: pause the SGLT2 inhibitor during vomiting, reduced oral intake or acute illness because of the risk of euglycemic ketoacidosis.

5.3 DPP-4 inhibitors

Sitagliptin is weight-neutral and has a low hypoglycemia risk. Usual dose is 100 mg daily; reduce to 50 mg daily when eGFR is 30 to 44 and to 25 mg daily when eGFR is below 30.

5.4 GLP-1 receptor agonists

A GLP-1 receptor agonist such as liraglutide is preferred when there is established atherosclerotic cardiovascular disease or when weight reduction is a priority. Gastrointestinal side effects are common during titration.

5.5 Sulfonylureas

Sulfonylureas such as glipizide are effective and inexpensive but carry a significant risk of hypoglycemia, particularly in older adults and in chronic kidney disease. Avoid them in patients with recurrent hypoglycemia.

5.6 Basal insulin initiation and titration

Start basal insulin when HbA1c is above 10%, when HbA1c remains above 9% despite two or three non-insulin agents, or when symptomatic hyperglycemia or catabolic features are present. The usual starting dose of insulin glargine is 10 units once nightly or 0.1 to 0.2 units per kg. Titrate by 2 units every 3 days until fasting glucose is between 80 and 130 mg/dL. Reduce the dose by 10 to 20% after

any hypoglycemia episode. Continue metformin when renal function allows.

5.7 Treatment intensification

If HbA1c is not at target after 3 months on the current regimen, intensify treatment. Do not delay intensification for more than 6 months when HbA1c remains above target (clinical inertia).

6. Inpatient Management

6.1 Inpatient glucose targets

For most hospitalised patients, maintain glucose between 140 and 180 mg/dL. Point-of-care glucose should be checked before meals and at bedtime, or every 4 to 6 hours when the patient is not eating.

6.2 Oral agents and insulin in hospital

Metformin and SGLT2 inhibitors are usually held on admission during acute illness. A scheduled basal-bolus insulin regimen is preferred; a correctional (sliding-scale) insulin regimen used alone is discouraged because it treats hyperglycemia after it occurs.

6.3 Hypoglycemia treatment

Level 1 hypoglycemia is glucose below 70 mg/dL and level 2 is below 54 mg/dL. Treat a conscious patient with 15 g of fast-acting carbohydrate and recheck after 15 minutes, repeating until glucose is above 70 mg/dL. An unconscious patient receives IV dextrose 10% or glucagon. Every episode must be documented and the insulin dose reviewed the same day.

6.4 Hyperglycemic emergencies

Suspected diabetic ketoacidosis (DKA) or hyperosmolar hyperglycemic state (HHS) must be managed with the Emergency Admission Procedure and the hyperglycemic emergency order set.

7. Discharge and Transitions of Care

Medication reconciliation is mandatory before discharge, with particular attention to insulin doses and any held oral agents. Patients admitted for hyperglycemia or whose insulin regimen changed must have a diabetes follow-up appointment within 7 to 14 days of discharge. Provide written sick-day rules and hypoglycemia education, and confirm that the patient has a working glucose meter. Patients identified as high risk of readmission should receive a telephone follow-up within 72 hours (see Discharge Planning and Transitions of Care Policy CF-POL-DC-07).

8. Document Control

Approved by the (fictional) CareFlow Clinical Guidelines Committee. Next review in 24 months. This document is synthetic and intended only for software demonstration.